COMP5331 Knowledge Discovery in Databases (Fall Semester 2026)
Homework 2
Deadline: 6 Nov, 2026 1:30pm
(Please hand in Canvas.)
Full Mark: 100 Marks

Coupon Instructions:

1. You can use a coupon to waive any question you want and obtain full marks for this question.
2. You can waive at most one question in each assignment.
3. You can also answer the question you will waive. We will also mark it but will give full marks to this question.
4. The coupon is non-transferrable. That is, the coupon with a unique ID can be used only by the student who obtained it in class.
5. Please scan the coupon (hardcopy) as an electronic copy
6. Please place this electronic copy of the coupon to your the submitted assignment (softcopy) on the first page
7. Please write down/type the question no. you want to waive on the coupon.

Q1 [20 Marks]

We are given l attributes, namely A_1, A_2, …, A_l, and n data points. Suppose that $n = 2^k$ where k is a positive integer ≥ 2. Given a data point x, we denote the value of attribute A for x to be x.A. Assume that two different data points have different attribute values for each attribute.

In the density-based subspace clustering, we learnt that each (grid) unit can be represented by p intervals if we consider p attributes only where $p \leq l$. "A_2=[1, 10], A_6=[21, 30]" is an example of the representation of a unit if we consider two attributes, A_2 and A_6. The number of attributes involved for this unit is 2. Besides, the length of the interval "A_i=[a, b]" is defined to be b-a where a and b are two real numbers. The volume of a unit is defined to be the product of the lengths of all intervals involved for this unit. Recall that in this density-based subspace clustering, we want to find all subspaces which contain dense units. Formally, given a subspace S in the result, there exists a unit such that the attributes involved for this unit are the attributes involved for S and this unit is dense.

However, in this density-based subspace clustering, the length of each interval along an attribute is fixed. Motivated by this observation, we want to define some intervals with varying lengths along an attribute according to the data distribution. In this way, we can define a unit based on these intervals.

According to the data, we can generate a set of intervals according to a function **Generate** (to be described). According to these intervals, we can define units. Before we describe function **Generate**, we give some concepts as follows.

We split a set G of points into two parts according to an attribute A such that there exist four numbers y_1, y_2, y_3, $y_4 \in$ A where

- $y_1 < y_2 < y_3 < y_4$
- $y_1 = \min_{x \in G} x.A$
- y_2 is equal to the attribute value of a data point in G
- y_3 is also equal to the attribute value of a data point in G
- $y_4 = \max_{x \in G} x.A$
- the total number of data points in G which attribute values on A are at most y_2 is equal to the total number of data points in G which attribute values on A are at least y_3
- there are no data points in G which attribute values on A are greater than y_2 and smaller than y_3

Let Left(G, A) be a set of data points in G which attribute values on A are at most y_2 (described above).
Let Right(G, A) be a set of data points in G which attribute values on A are at least y_3 (described above).
Let SplitValue(G, A) be $[(y_2-y_1)+(y_4-y_3)]/2$ (where y_1, y_2, y_3 and y_4 were described above).

Now, we define a function **Generate** which takes a set G of data points as an input and outputs a set of intervals.

Function **Generate**(G)

if $|G| \leq 4$
    $X \leftarrow \varnothing$
    for each $i \in [1, l]$ do
        $y_1 \leftarrow \min_{x \in G} x.A_i$
        $y_4 \leftarrow \max_{x \in G} x.A_i$
        $interval_i \leftarrow [y_1, y_4]$
        $X \leftarrow X \cup$ "$A_i = interval_i$"
else
    $A_i \leftarrow$ the attribute A which has the greatest value of SplitValue(G, A) (among all attributes)
    $G_{left} \leftarrow$ Left(G, A_i)
    $G_{right} \leftarrow$ Right(G, A_i)
    $X_{left} \leftarrow$ **Generate**(G_{left})
    $X_{right} \leftarrow$ **Generate**(G_{right})
    $X \leftarrow X_{left} \cup X_{right}$
return X

(a) Is it possible that the intervals along a single attribute which are generated by function **Generate** are overlapping? Why?
(b) Under this new definition of intervals, we have a new definition of "dense" as follows.
Suppose that we define that a unit is dense if this unit contains at least 4 data points and there exists an interval of this unit where the length of this interval is smaller than or equal to a support threshold t where t is a non-negative real number and a user parameter. Can we still adopt the Apriori-like algorithm for finding all subsapces containing dense units? If yes, please describe how to adopt the algorithm. Otherwise, please give reasons why it cannot be adopted.
(c) This part is independent of part (b).
Under this new definition of intervals, we have a new definition of "dense" as follows.
Suppose that we define that a unit is dense if this unit contains at least 4 data points and the volume of this unit is smaller than or equal to t^s where t is a non-negative real number and a user parameter, and s is the number of attributes involved for this unit. Can we still adopt the Apriori-like algorithm for finding all subsapces containing dense units? If yes, please describe how to adopt the algorithm. Otherwise, please give reasons why it cannot be adopted.
(d) What are the advantages of using this approach for subspace clustering compared with the density-based subspace clustering you learnt in class?

Q2 [20 Marks]

(a) Consider two random variables X and Y with the following probabilistic table.

X \ Y	1	2	3
1	0	1/8	1/8
2	1/2	0	1/8
3	1/8	0	0

(i) Calculate the conditional entropy of H(X|Y) by using the original definition of the conditional entropy.

(ii) Calculate H(X|Y) as

$$- \sum_{x \in A} \sum_{y \in B} p(x, y) \log p(x|y)$$

where A = {1, 2, 3} and B = {1, 2, 3}.

(b) Consider the four 2-dimensional data points:

(2, 2), (4, 4), (1, 5) and (5, 1)

We can make use of the KL-Transform to find a transformed subspace containing a cluster. Let L be the total number of dimensions in the original space and K be the total number of dimensions in the projected subspace. Please illustrate the KL-transform technique with the above example when L=2 and K=1. Also, shows your results in a diagram indicating all the points in the original coordinate system and the two transformed axes.

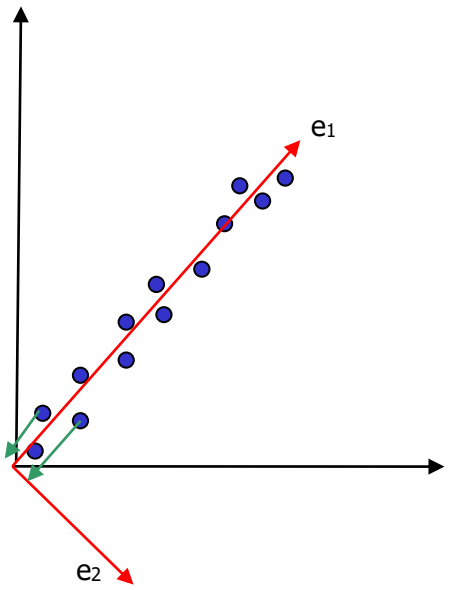


Figure 1

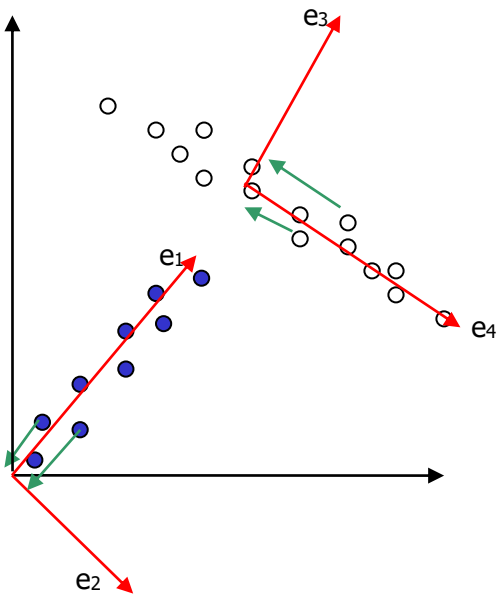


Figure 2

(c) The KL-transform technique discussed in class only focuses on finding a single cluster along a "transformed" subspace. In many cases, there are multiple clusters along multiple "transformed" subspaces. In the example shown in Figure 1 containing a single cluster along a "transformed" subspace, the KL-transform technique discussed in class can find the cluster projected on the new axis e_2. However, in the example shown in Figure 2 containing two clusters (one cluster containing all blue points along the new axis e_2 and the other cluster containing all white points along the new axis e_3), the original KL-transform technique discussed in class cannot find these two clusters. Suppose that we know that there are n clusters (along the "transformed" subspaces). Please extend the KL-transform technique so that we can find multiple clusters along multiple "transformed" subspaces. Please write the pseudo-code and explain it.

Q3 [20 Marks]

The following shows a history of customers with their incomes, an attribute called "Have_iPhone17" and another attribute called "Have_iPad" . "Have_iPhone17" denotes whether a customer has an iPhone17 and "Have_iPad" denotes whether a customer has an iPad. We also denote whether they will buy an iPhone17Duo or not in the last column.

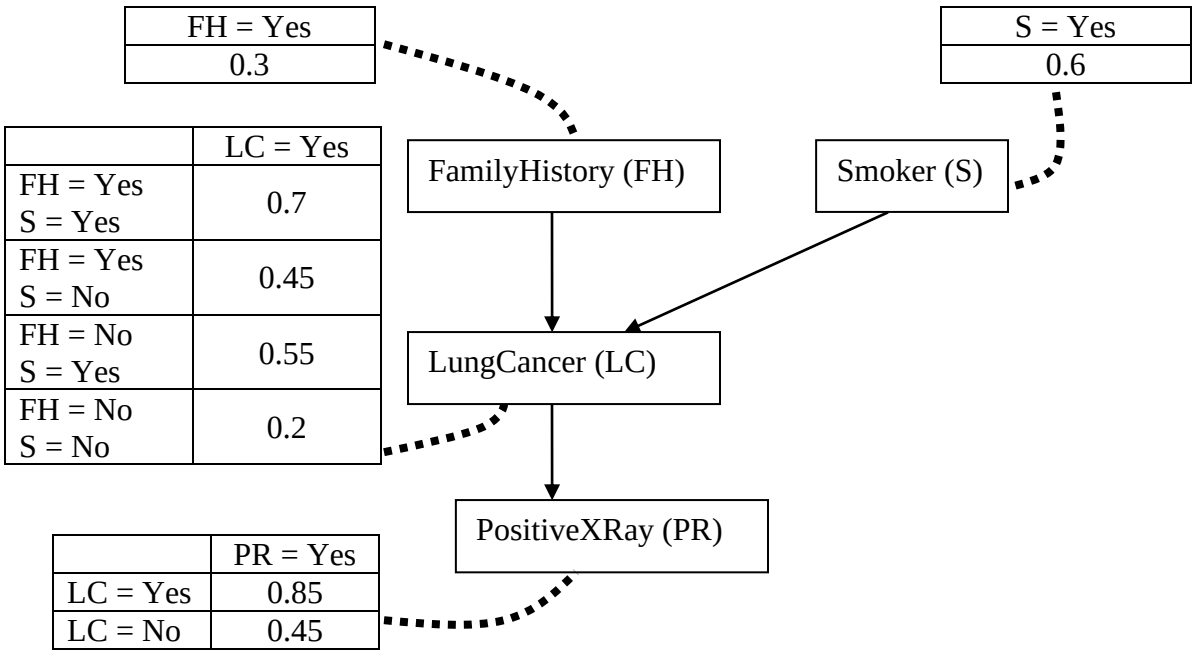
No.	Income	Have_iPhone17	Have_iPad	Buy_iPhone17Duo
1	high	yes	yes	yes
2	high	no	yes	yes
3	medium	yes	no	yes
4	high	no	no	yes
5	medium	yes	no	no
6	medium	yes	no	no
7	medium	no	no	no
8	medium	no	no	no

(a) We want to train a CART decision tree classifier to predict whether a new customer will buy an iPhone17Duo or not. We define the value of attribute Buy_iPhone17Duo is the *label* of a record.
   (i) Please find a CART decision tree according to the above example. In the decision tree, whenever we process a node containing at most 3 records, we stop to process this node for splitting.
   (ii) Consider a new customer whose income is medium and he has both an iPhone17 and iPad. Please predict whether this new customer will buy an iPhone17Duo or not.

(b) What is the difference between the C4.5 decision tree and the ID3 decision tree? Why is there a difference?

Q4 [20 Marks]

We have the following Bayesian Belief Network.



	LC = Yes
FH = Yes S = Yes	0.7
FH = Yes S = No	0.45
FH = No S = Yes	0.55
FH = No S = No	0.2

	PR = Yes
LC = Yes	0.85
LC = No	0.45

Suppose that there is a new person. We know that

(1) he has his family history
(2) he is a non-smoker
(3) his result of X-Ray is positive

We would like to know whether he is likely to have Lung Cancer.

Family History	**Smoker**	**PositiveXRay**	**Lung Cancer**
Yes	No	Yes	?

(a) Please use Bayesian Belief Network classifier with the use of Bayesian Belief Network to predict whether he is likely to have Lung Cancer.

(b) Although Bayesian Belief Network classifier does not have an independent assumption among all attributes (compared with the naïve Bayesian classifier), what are the disadvantages of this classifier?

Q5 [20 Marks]

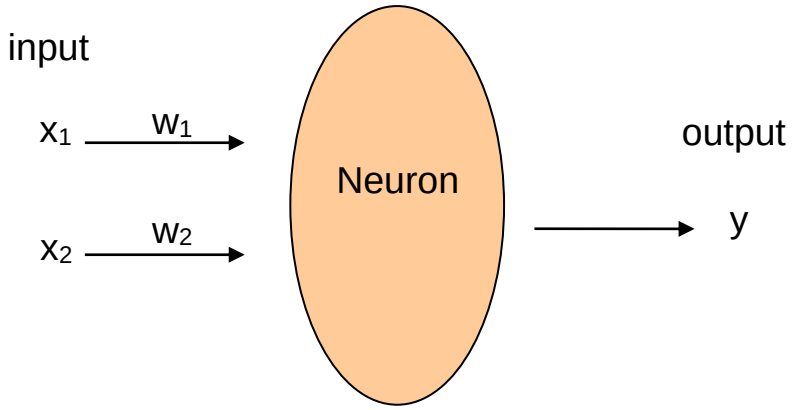
Consider the following table with three attributes where "No. of Computers" and "No. of Mobile Devices" are input attributes and "Buy_Norton360" is the target attribute. Each tuple corresponds to a customer.

No. of Computers	No. of Mobile Devices	Buy_Norton360
2	0	No
0	2	No
4	2	Yes
2	4	Yes

(a) Suppose that there is a new customer with "No. of Computers" = 1 and "No. of Mobile Devices" = 0. Please predict whether this customer will buy Norton360 or not by a 3-nearest neighbor classifier. You just need to write down the answer without any explanation.

(b) (i) Rewrite the above table such that values "Yes" and "No" in attribute "Buy_Norton360" are mapped to values 1 and -1, respectively.
(ii) Formulate this SVM problem by a quadratic programming. Please list the objective function and all constraints in this quadratic function.

(c) (i) Rewrite the above table such that values "Yes" and "No" in attribute "Buy_Norton360" are mapped to values 1 and 0, respectively.
(ii) Consider a neural network containing a single neuron where x_1 = "No. of Computers", x_2 = "No. of Mobile Devices" and y = "Buy_Norton360".



Initially, we set the values of w_1, w_2 and b to be 0.3 where b is a bias value in the neuron.

Suppose the learning rate is denoted by α. Let $\alpha = 0.6$.
Suppose we adopt the threshold function as an activation function.
Please try to train the neural network with the following five instances in the given sequence based on the answer in (c)(i).

1. $(x_1, x_2) = (2, 0)$
2. $(x_1, x_2) = (0, 2)$
3. $(x_1, x_2) = (4, 2)$
4. $(x_1, x_2) = (2, 4)$
5. $(x_1, x_2) = (2, 0)$

What are the final values of w_1, w_2 and b after these five instances are read?